

Network storage - In an aviation R&D centre, different engineering/aerospace users may generate large volumes of:

Engineering and design data

Simulation/testing data

Aircraft/component research data

Technical documents

Application/system data

The NetApp infrastructure provides centralised storage and access to this data over the organisation's network.

The asset is a replacement of the existing storage infrastructure, which was nearing the end of its 3-year useful life.

Opal-rt - OPAL-RT is a specialised real-time simulation/testing machine/computing device used by engineers to model and test how an aircraft electrical power systems will behave under different conditions, it is capable of performing the high-speed processing required for the simulations.

This has been installed in current year keeping in mind the increased requirement due to project expansion.

Warm fuel coking rig - The Warm Fuel Coking Rig allows engineers to circulate aviation turbine fuel at controlled temperature, pressure and flow conditions and study its behaviour, including its tendency to form deposits/coke under warm operating conditions

The rig comprises specialised pumps, heaters, pressure regulators, analysers, chillers, flow meters, fuel tanks and associated tubing/structural components.

This is a new project for which a new lab has been created and this was the asset procured in that lab

SEM with oxford EDS attachment

The SEM (scanning electron microscope) with Oxford EDS (Energy Dispersive Spectroscopy) is specialised material-analysis equipment used at the Jfwtc R&D centre to examine engine components such as turbine blades at microscopic levels. The SEM provides detailed microscopic images, while the EDS attachment identifies the elemental composition of the material. It is primarily used for investigating material degradation, defects and creep-related changes in engine components. The asset was procured as a replacement for the existing SEM infrastructure.

Miniature robotics system lab -

Miniature robotic inspection system comprising robotic arm, optical microscopy/camera system and related equipment, used for automated inspection and analysis of aircraft engine blades as part of R&D activities. This is a new lab that has been created in response to project expansion and increased requirements

Ductalke split ac at Odyssey building -

Replacement and upgradation of the existing AC infrastructure at Odyssey Building, as the existing system was nearing the end of its useful life (10–15 years). The obsolete split AC units were replaced with a new VRF (Variable Refrigerant Flow) system, which enables centralised and more efficient cooling with independent temperature control across different areas.

Vrf coil for ph4 acu - Replacement/upgradation of the existing HVAC infrastructure at Galileo Building, with VRF coils integrated with AHUs (air handling units) as part of the new VRF-based system. The existing AC infrastructure was nearing the end of its useful life and was replaced with upgraded VRF technology.

Xray system -

Replacement of the existing 450 kV industrial X-ray system used for non-destructive testing of aerospace and engine components at the R&D centre. The system enables engineers to examine the internal structure of components and identify defects such as cracks, voids, porosity and other abnormalities without physically damaging the components. The new system replaces the existing equipment and supports R&D and component inspection activities.